

Acoustic Horizons: A multimodel approach to blind navigation with scene optimised YOLOv8 Object Detection, Enhanced by Bone Conduction Technology

Rithvik Grandhi

*Dept. of Computer Science and Engineering
PES University
Bengaluru, India
grandhirithwik@gmail.com*

Rishikesh Sastri

*Dept. of Computer Science and Engineering
PES University
Bengaluru, India
rishi95sastri@gmail.com*

Revathi GP

*Dept. of Computer Science and Engineering
PES University
Bengaluru, India
revathigp@pes.edu*

Rohan C

*Dept. of Computer Science and Engineering
PES University
Bengaluru, India
chandrashekar.rohans@gmail.com*

Neil William Fernandes

*Dept. of Computer Science and Engineering
PES University
Bengaluru, India
neilpinef@gmail.com*

Abstract—This paper introduces an innovative wearable lightweight, wireless headset designed to assist visually impaired individuals in navigating their environment more effectively and safely. The device aims to address common challenges faced by this demographic, including obstacle detection and spatial orientation, thereby enhancing their ability to move independently and confidently in various settings using 2 low powered esp32 camera modules which provide live, low-latency video feeds to the respective YOLOv8 model based on the scene which act as an extremely low cost alternative to AI cameras which are orders of magnitude more expensive.

Utilizing advanced sensory and computational technologies, the headset provides real-time feedback and guidance, enabling users to identify and avoid potential hazards in their path. This research outlines the development process, technical specifications, and the results of preliminary user testing, demonstrating the headset's potential to significantly improve the quality of life for visually impaired individuals by offering a practical solution for basic navigation and obstacle avoidance.

Index Terms—Computer Vision, Object detection, YOLOv8, Bone Conduction, ESP32, Places365, Bone conduction

I. INTRODUCTION

As per the global data on visual impairment by WHO (2010), 285 million people are reported to be blind in the world, out of this a staggering 39 million are completely blind with irreversible damages to eye [8]

Visually impaired individuals often face significant challenges in navigating through their environment safely and independently. Traditional aids like canes and guide dogs [9] provide limited information about distant obstacles and the surrounding environment [10] [15]. Other alternatives which can be easier to use such as a system using Ultrasonic sensor for the detection of an obstacle is developed.[11] In this system an overcoat is designed with two ultrasonic sensors using which the presence is detected at a distance which ranges from 3cm to 3m and the field of view is limited to 60°[12]. Once when an obstacle is detected this overcoat produces a whole body vibrating effect in order to indicate obstacles presence [13]

Recent advancements in technology have opened new possibilities for creating more effective navigational aids. This paper introduces a novel headset that leverages cutting-edge computer vision and audio feedback technology to aid visually impaired individuals in navigating their surroundings.

II. SYSTEM DESIGN

A. Hardware Components

As seen in Fig. 1 The system's core components include two ESP32 camera modules, strategically placed on either side of a headband to provide a wide field of view and depth perception. These cameras feed live video data to a processing unit running

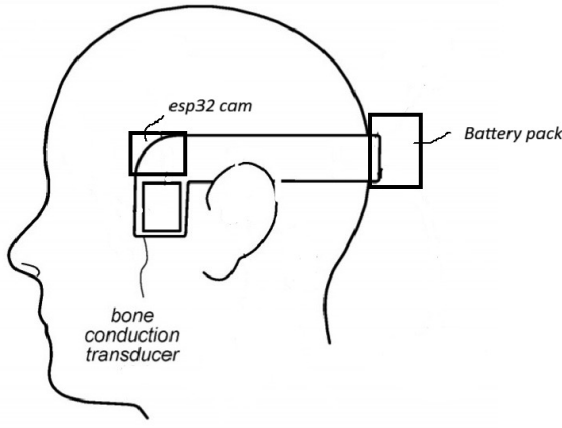


Fig. 1. Wireframe of the proposed design.

the state-of-the-art YOLOv8 object detection algorithm. The binaural audio feedback mechanism is implemented through bone conduction headphones, which are integrated into the headset.

B. Software and Algorithms

At the heart of our system lies the YOLOv8 model, known for its exceptional speed and accuracy in object detection. This model is specifically optimised to process stereo images captured by ESP32 cameras, facilitating the real-time identification and localization of obstacles. The effectiveness of the YOLOv8 model in detecting a wide range of objects is vital for the system's capability to navigate diverse environments.

In addition to object detection, our system incorporates a separate model trained on the Places365 dataset for scene recognition. This model plays a pivotal role in identifying the user's current environment, providing context-sensitive information that is crucial for accurate navigation assistance. By understanding the type of environment the user is in, the system can adjust its processing and feedback mechanisms more effectively.

Our approach includes the creation of custom datasets, which are tailored to specific scenes and environments encountered by users. These specialized datasets play a key role in enhancing the efficiency of our object detection model. Multiple YOLOv8 models trained on these custom datasets can switch seamlessly based on the scene which the above mentioned model passes the scene information every set interval of time to reduce the computational load. By minimizing the dataset size and using different models for each scene, we achieve faster detection speeds, much better accuracy compared to using a single model and significantly reduce the power consumption of the processing unit, an essential aspect for portable, battery-powered devices. This includes consideration of contours around the objects if it is indoors as shown in Fig. 3 which can provide much more accurate avoidance and indoor spaces can be limited in size whereas the outdoor model can use just bounds as shown in Fig. 4 which can save on a lot of computation.

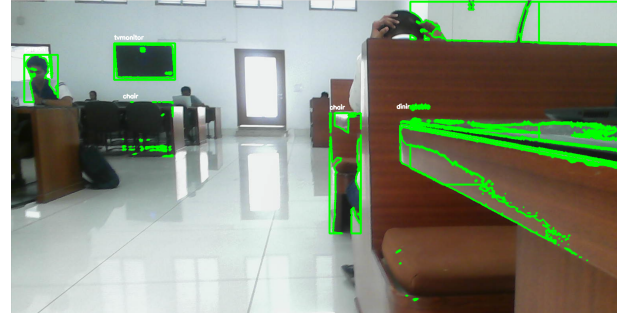


Fig. 2. Object detection indoors with contours for better accuracy

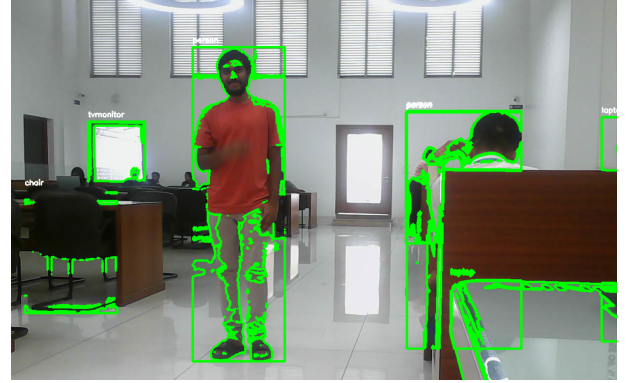


Fig. 3. Highlighting the edge detection showing the edge of a big table and the base of a person



Fig. 4. Object detection outdoor with only bounding boxes

Depth estimation forms a crucial aspect of our system, utilizing sophisticated methods to analyze stereo vision data captured by dual cameras. This process involves a two-pronged approach to calculate distances accurately. Firstly, we determine the centroid of the bounding box, identified around each detected object. This centroid serves as a focal point for measuring the Euclidean distance between the object and the camera, providing a fundamental distance estimation.

Secondly, we enhance this basic measurement with the aid of disparity maps generated from the stereo images. Disparity maps offer a detailed representation of depth by highlighting

the differences in images captured from slightly different viewpoints, as is the case with our dual camera setup. By integrating the disparity map data with the initial Euclidean distance calculations, we are able to achieve a highly precise estimation of the distance between objects and the user.

The combination of these techniques, the centroid-based Euclidean distance calculation and disparity map analysis ensures the depth measurements are both accurate and reliable. This can be seen in Fig. 5. This level of precision in depth estimation is vital for the system to provide effective and dependable navigation assistance. It allows for nuanced and responsive feedback, tailored to the specific spatial arrangement of the user's environment, thereby enhancing the safety and navigational capabilities of visually impaired individuals.

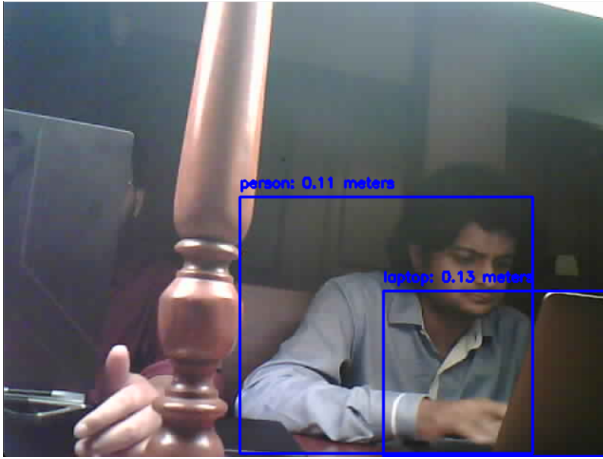


Fig. 5. Depth estimation

The integration of these elements—advanced object detection, contextual scene recognition, optimised custom datasets, and precise depth estimation—culminates in a robust system. This integration ensures not only the accuracy of the data acquired but also the efficiency and reliability of the system. By effectively processing and interpreting complex visual information, our system provides a highly effective navigation aid for visually impaired individuals, enhancing their independence and mobility.

C. Obstacle Detection and Depth Perception

The system excels in identifying potential hazards in the user's path and precisely estimating their distance, a key factor for calculating the intensity and directionality of the auditory feedback. To achieve this, the system employs two distinct methods for distance calculation: the distance formula and disparity map analysis.

Firstly, the distance formula method utilizes the spatial coordinates of objects detected by the cameras. By measuring the actual size of an object and comparing it with its perceived size in the image, the system calculates the distance based on geometric principles. This approach is particularly effective for well-defined objects with known dimensions, providing a straightforward and reliable means of distance estimation.

Secondly, the disparity map analysis is a core feature of our stereo vision system. This method involves comparing the differences (or disparities) in the images captured by the two cameras. Objects closer to the cameras will have more significant disparities between the two images, whereas distant objects will have less. By analyzing these disparities, the system can construct a depth map of the scene, accurately gauging the distance of various objects. This technique is invaluable for perceiving depth in complex environments where objects may not have predefined sizes or shapes.

Together, these dual methods of distance calculation provide a comprehensive and nuanced understanding of the environment. The distance formula offers a straightforward estimation for known objects, while the disparity map analysis fills in the gaps, allowing for accurate depth perception in more dynamically varying and unpredictable environments. This dual approach ensures that the system can adapt to a wide range of scenarios, providing reliable and intuitive feedback for navigation.

D. Auditory Feedback System

In a research conducted, it was found that the blind participants strongly favor haptics over audio. White cane users were accustomed to perceive short range navigation information from the haptic channel, and therefore found it intuitive to use as shown by Martinez et al [1].

Our innovative auditory feedback system employs bone conduction technology to deliver audio cues without impeding the user's ability to hear environmental sounds. This feature is crucial for the safety of visually impaired users, as it allows them to remain alert to their surroundings while receiving navigational assistance.

The sophistication of the system lies in its nuanced approach to sound conveyance. To guide a visually impaired person using binaural reproduction, the accuracy of sound localization performance should be as high as possible to guarantee the spatial resolution of every direction of the clock positions used by visually impaired people for direction recognition (every 30°, twelve divisions of 360° in the horizontal plane) et al Takumi Asakura [2] While the primary indicator of an obstacle's location is the ear towards which the sound is directed (left or right), the system also finely adjusts the intensity of the sound in both ears which is to reflect the obstacle's precise position. This method is akin to advanced surround sound systems, where the sound's spatial characteristics are key to creating a three-dimensional auditory experience. This fine adjustment differs from person to person and can be calculated using 3 types of pink noises, of which the Pink1 (1/6 octave Pink noise signal with a nominal center frequency at 1000 Hz) had a greater impact in creating the sensation of two sound images at different points in space as found by et al M. Daraban [3] The system communicates the distance to obstacles to the user through a carefully designed auditory feedback mechanism. The frequency of the emitted beeps is adjusted to be logarithmically proportional to the calculated distance to the obstacles. This approach aligns with the natural

human sensitivity to loudness versus intensity, ensuring that the perceived change in beep frequency intuitively corresponds to changes in distance. In essence, as an obstacle gets closer, the frequency of the beeps increases quicker in a manner that is naturally perceptible to the user, allowing for an accurate and instinctive understanding of the surrounding environment. This method of conveying distance through sound is not only effective but also aligns with innate human auditory processing, enhancing the usability and effectiveness of the system. Another advantage of using binaural system is that we need not be concerned with the shape of the person's ear which in a traditional sound system would matter as the shape of the ear is what lets a person sense the direction of the sound [14].

$$f = k \cdot \log(d + 1)$$

where

- f is the frequency of the beeps,
- d is the calculated distance to the object,
- k is the distance constant
- * +1 so log is never 0

For example, if an obstacle is detected to the left of the user, the beep sound will be more intense in the left ear. However, the right ear will also receive this sound at a lower intensity. This gradation in intensity across both ears provides a more accurate perception of the obstacle's location, mirroring the way humans naturally perceive sound in a three-dimensional space. It allows users to discern not just the side of the obstacle but also its angle and relative position with greater accuracy.

By integrating this level of detail into the auditory feedback, the system enables users to navigate more effectively, offering a richer and more informative audio experience. This approach significantly enhances the user's spatial awareness, contributing to safer and more confident navigation in various environments.

III. PRELIMINARY TESTING AND RESULTS

Initial testing with visually impaired volunteers showed promising results. Users reported an enhanced sense of their surroundings and could avoid obstacles more effectively than with traditional aids. Further quantitative data on detection range, accuracy, and user response times are currently being compiled for a more comprehensive analysis.

IV. CONCLUSION AND FUTURE WORK

This paper presented a novel guidance system for the visually impaired, demonstrating the potential of integrating computer vision and auditory feedback in a wearable format. Our proposed system uniquely combines the YOLOv8 model with bone conduction technology, providing an innovative solution for obstacle detection and navigation.

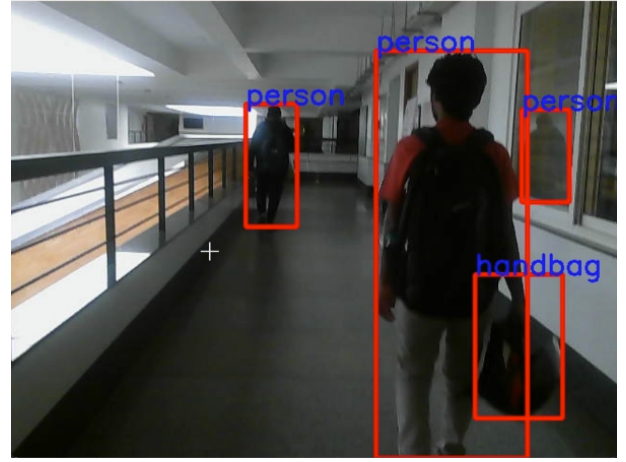


Fig. 6. Navigating a narrow crowded hallway

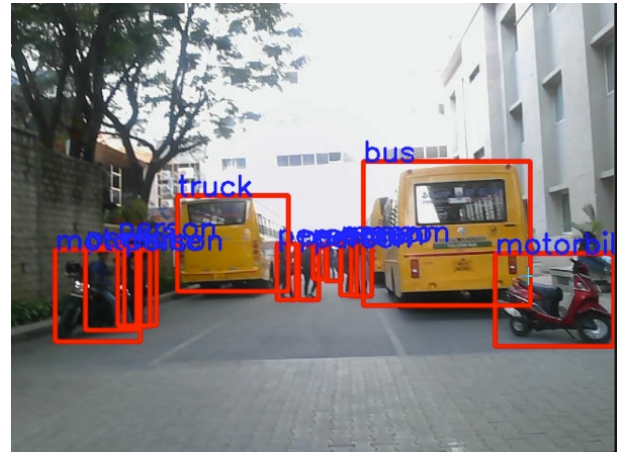


Fig. 7. Navigating a crowded road with a lot of vehicles

The preliminary results are promising, indicating an enhanced spatial awareness and obstacle avoidance capability for visually impaired individuals.

The past few decades have seen a rapid increase in micro-processor power and a consequent drop in the cost and power consumption per floating point operations per second [6]. While most of the headlines are grabbed by microprocessor clock-speed champions, a quiet progress of low budget, power autonomous devices in industry and in our daily routines is riding on a wave of inexpensive, power efficient microcontrollers [7] [5].

Therefore, the current prototype requires further refinement for everyday practical use. Future work will focus on miniaturizing the design, improving battery life, and enhancing the comfort and ergonomics of the headset.

Extensive user trials are planned to validate the effectiveness of the system in real-world scenarios. We aim to gather comprehensive feedback from visually impaired individuals to understand better their needs and the system's impact on their daily navigation and independence.



Fig. 8. Front view of the prototype



Fig. 9. Top view of the prototype

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REFERENCES

- [1] Martinez, M., Constantinescu, A., Schauerte, B., Koester, D., Stiefelha-gen, R. (2014). Cognitive Evaluation of Haptic and Audio Feedback in Short Range Navigation Tasks. In: Miesenberger, K., Fels, D., Archambault, D., Peñáz, P., Zagler, W. (eds) Computers Helping People with Special Needs. ICCHP 2014. Lecture Notes in Computer Science, vol 8548. Springer, Cham. https://doi.org/10.1007/978-3-319-08599-9_20.
- [2] Asakura, T. Bone Conduction Auditory Navigation Device for Blind People. Appl. Sci. 2021, 11, 3356. <https://doi.org/10.3390/app11083356>.
- [3] M. Daraban, K. Lengyel, A. Faluvegi and C. Amariei, "Real-time Spatialized Sound Generator on Embedded Wearable Platform," 2021 IEEE 27th International Symposium for Design and Technology in Electronic Packaging (SIITME), Timisoara, Romania, 2021, pp. 71-76, doi: 10.1109/SIITME53254.2021.9663671.
- [4]
- [5] B. Salonikidou, D. Savvas, G. Diamantis and A. Astaras, "Development and evaluation of an open source wearable navigation aid for visually impaired users (CYCLOPS)," 2012 IEEE 12th International Conference on Bioinformatics & Bioengineering (BIBE), Larnaca, Cyprus, 2012, pp. 115-120, doi: 10.1109/BIBE.2012.6399659.
- [6] O. Silven and K. Jyrkka, "Observations on power-efficiency trends in mobile communication devices," EURASIP J. Embedded Syst., v. 2007:1, pp. 7-17, jan 2007.
- [7] M. Frankowiak, R. Grosvenor, et al, "A review of the evolution of microcontroller-based machine and process monitoring," Intl J of Machine Tools and Manufacture, vol.45:4-5, pp. 573-82, apr 2005.
- [8] K. Viswanathan and S. Sengupta, "Blind navigation proposal using SONAR," 2015 IEEE Int. Conf. Comput. Graph. Vis. Inf. Secur. CGVIS 2015, pp. 151-156, 2016.
- [9] A. Kulkarni, A. Wang, L. Urbina, A. Steinfeld, and B. Dias, "Robotic assistance in indoor navigation for people who are blind," ACM/IEEE Int. Conf. Human-Robot Interact., vol. 2016-April, pp. 461-462, 2016.
- [10] K. K. O. Sharma, N. Jain, A. K. Bhargav and S. B. Bhanu Prashanth, "Design development of a bone conduction based navigation aid for the visually impaired," 2017 2nd IEEE International Conference on Recent Trends in Electronics, Information Communication Technology (RTEICT), Bangalore, India, 2017, pp. 1699-1702, doi: 10.1109/RTE-ICT.2017.8256889.
- [11] H. Archana, C. Chellaram and M. Rajalakshmi, "Obstacle detection for visually impaired patients," 2014 International Conference on Science Engineering and Management Research (ICSEMR), Chennai, India, 2014, pp. 1-3, doi: 10.1109/ICSEMR.2014.7043565.
- [12] Sylvain Cardin, Daniel Thalmann and Frederic "Wearable Obstacle Detection System for visually impaired People" Vexo, Virtual Reality Laboratory (VRLab).
- [13] A.Panuccio "A Multimodal Electronic Travel Aid Device" ICMI 02: in proceedings of the 4th IEEE International Conference on Multimodal Interfaces, IEEE Computer society Washington DC, USA, pp 39, 2002.
- [14] J. Xiao, K. Ramdath, M. Losilevish, D. Singh and A. Tsakas, "A low cost outdoor assistive navigation system for blind people," 2013 IEEE 8th Conference on Industrial Electronics and Applications (ICIEA), Melbourne, VIC, Australia, 2013, pp. 828-833, doi: 10.1109/ICIEA.2013.6566481.
- [15] "Challenges That Blind People Face", .livestrong.com/article/241936-challenges-that-blind-people-face/